Harmonic oscillator 1

Let $V(x)$ be the potential energy for a harmonic oscillator.

$$V(x) = \frac{1}{2} m\omega^2 x^2$$

We seek $\psi(x,t)$ that solves the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x,t) + V(x)\psi(x,t)$$

The solution for $n = 0, 1, 2, \ldots$ is

$$\psi_n(x,t) = C_n \exp\left(-\frac{iE_n t}{\hbar}\right) \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^n}{dx^n} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

where C_n is the normalization constant

$$C_n = \frac{(-1)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left(\frac{\hbar}{m\omega}\right)^{n/2}$$

and E_n is the energy level

$$E_n = \hbar\omega \left(n + \tfrac{1}{2}\right)$$

Taking the time derivative of $\psi_n(x,t)$ we find

$$i\hbar \frac{\partial}{\partial t}\psi_n(x,t) = E_n \psi_n(x,t)$$

Hence by the Schrodinger equation

$$E_n \psi_n(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi_n(x,t) + V(x)\psi_n(x,t)$$

Noting that the factor $\exp(-iE_n t/\hbar)$ cancels we also have

$$E_n \phi_n(x) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\phi_n(x) + V(x)\phi_n(x)$$

where

$$\phi_n(x) = C_n \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^n}{dx^n} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

and

$$\psi_n(x,t) = \exp\left(-\frac{iE_n t}{\hbar}\right) \phi_n(x)$$

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